

Project Initialization and Planning Phase

Date	03 July 2026
Team ID	XXXXXX
Project Name	Smart Lender - AI Powered Loan Eligibility Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal aims to transform the loan approval process using machine learning, improving speed, consistency and accuracy of eligibility decisions. It tackles the inefficiencies of manual review, promising faster operations, reduced default risk and a better applicant experience through an instant, data-driven eligibility prediction.

Project Overview

Objective	Develop an AI system that accurately predicts loan eligibility from applicant data.
Scope	Predict loan approval status • Display confidence and risk level • Web-based application

Problem Statement

Description	Manual loan review is slow and inconsistent, causing delays for applicants and risk exposure for lenders.
Impact	Faster decisions • Consistent approvals • Reduced manual workload

Proposed Solution

Approach	Train and compare multiple ML classifiers (Decision Tree, Random Forest, KNN, XGBoost) on historical loan application data, then deploy the best performing model (KNN, 84.55% testing accuracy) through a Flask web application.
Key Features	Real-time loan prediction • Confidence score • Risk classification • Applicant summary • Responsive UI

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU specifications, number of cores	Standard multi-core CPU (no GPU required for KNN/tree models)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models and logs	Under 1 GB (dataset + serialized models)
Frameworks	Python web framework	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, joblib, xgboost
Development Environment	IDE / Notebook	Jupyter Notebook, VS Code

Resource Type	Description	Specification / Allocation
Data	Source, size, format	Kaggle Loan Prediction dataset, 614 records, CSV